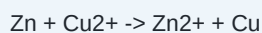


Learning target and prerequisites

GOVERNED REACTION



DECLARATIVE EXPLANATION

Learn to prove Redox roles from reaction evidence. Prerequisites: species identity, elemental oxidation state 0, ionic charge in $\text{Cu}^{2+}/\text{Zn}^{2+}$, and the meaning of increase/decrease.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

1. Track identity before labels

TRACK THE SAME ELEMENT BEFORE COMPARING



DECLARATIVE EXPLANATION

Locate Zn on both sides and Cu on both sides. Do not begin with oxidation, reduction or agent words. Identity is the anchor that makes every later comparison legal.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

2. What an oxidation-state lane does

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

DECLARATIVE EXPLANATION

The lane is a bookkeeping representation for the same element before and after the reaction. In this governed example Zn is 0 then +2; Cu is +2 then 0.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

3. Work the Zn lane

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

DECLARATIVE EXPLANATION

Zn starts as elemental Zn, oxidation state 0, and appears as Zn^{2+} , oxidation state +2. The number increases by 2; therefore Zn is oxidised.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

4. Work the Cu lane

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

DECLARATIVE EXPLANATION

Cu begins in Cu^{2+} with oxidation state +2 and ends as elemental Cu with oxidation state 0. The number decreases by 2; therefore Cu^{2+} is reduced.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

5. General direction rule

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

DECLARATIVE EXPLANATION

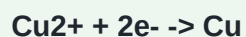
Increase -> oxidation. Decrease -> reduction. This is the concept-level rule; it should still work when the reagent names change.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

6. Translate Zn to electrons

ELECTRON LEDGER



2 electrons lost = 2 electrons gained

DECLARATIVE EXPLANATION

$\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^{-}$. Electrons appear on the product side because Zn loses them.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

7. Translate Cu²⁺ to electrons

ELECTRON LEDGER



2 electrons lost = 2 electrons gained

DECLARATIVE EXPLANATION

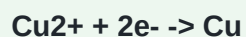
$\text{Cu}^{2+} + 2\text{e}^{-} \rightarrow \text{Cu}$. Electrons appear on the reactant side because Cu^{2+} gains them.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

8. Close the electron ledger

ELECTRON LEDGER



2 electrons lost = 2 electrons gained

DECLARATIVE EXPLANATION

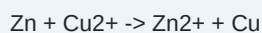
The same two electrons lost by Zn are gained by Cu²⁺. If the loss/gain counts disagree, the proof is internally inconsistent.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

9. Derive the Zn role

GOVERNED REACTION



DECLARATIVE EXPLANATION

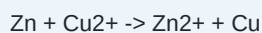
Prove SELF first: Zn is oxidised / loses electrons. Then OTHER effect: those electrons reduce Cu^{2+} . Conclusion: Zn is the reducing agent.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

10. Derive the Cu²⁺ role

GOVERNED REACTION



DECLARATIVE EXPLANATION

Prove SELF first: Cu²⁺ is reduced / gains electrons. Its acceptance of electrons makes oxidation of Zn possible. Conclusion: Cu²⁺ is the oxidising agent.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

11. Misconception contrast

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

DECLARATIVE EXPLANATION

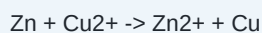
Wrong: 'Zn is reducing agent because Zn gets reduced.' Correct: 'Zn is oxidised; its electron loss reduces Cu²⁺; therefore Zn is the reducing agent.' The wrong statement reverses the proved self-change.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

12. Fully worked proof

GOVERNED REACTION



DECLARATIVE EXPLANATION

Zn 0 $\rightarrow$ +2, so Zn is oxidised. $\text{Zn} \rightarrow \text{Zn}^{2+} + 2\text{e}^-$. Cu +2 $\rightarrow$ 0, so Cu^{2+} is reduced. $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$. The two-electron ledger closes. Hence Zn is the reducing agent and Cu^{2+} is the oxidising agent.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

13. Completion / fading

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

DECLARATIVE EXPLANATION

Complete: Zn: 0 -> __ ; direction = __ ; process = __ ; electron statement = __ ; role = __. Then do the Cu line with fewer prompts.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.

14. Independent reconstruction

OXIDATION-STATE LANES

Zn: 0 -> +2 (increase)

Cu: +2 -> 0 (decrease)

DECLARATIVE EXPLANATION

Close the notes. Rebuild the complete proof from the equation only. Final check: same element tracked, state direction correct, electron loss = gain, agent label derived after self-change.

SELF-CHECK

Explain why the step is chemically legal before moving on. If the statement cannot be tied to the same element or the closed electron ledger, revisit the preceding page.